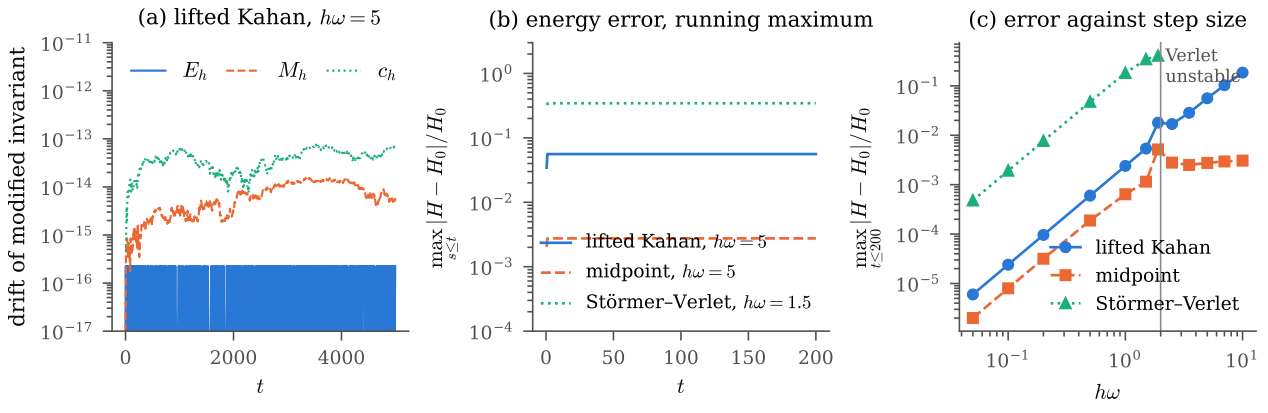
(a) lifted Kahan, $h\omega = 5$
drift of modified invariant
E_h
M_h
c_h
10^{-11}
10^{-12}
10^{-13}
10^{-14}
10^{-15}
10^{-16}
10^{-17}
0
2000
4000
t
(b) energy error, running maximum
$\max_{s \le t} |H - H_0| / H_0$
10^{0}
10^{-1}
10^{-2}
10^{-3}
10^{-4}
lifted Kahan, $h\omega = 5$
midpoint, $h\omega = 5$
Störmer–Verlet, $h\omega = 1.5$
0
50
100
150
200
t
(c) error against step size
$\max_{t \le 200} |H - H_0| / H_0$
Verlet unstable
10^{-1}
10^{-2}
10^{-3}
10^{-4}
10^{-5}
lifted Kahan
midpoint
Störmer–Verlet
10^{-1}
10^{0}
10^{1}
$h\omega$